Week 1: BattleOps Modules

Core Infra Foundations & Cloud-Native Deep Dive

MODULE 1: OS + System Fundamentals

What to Study

- Linux boot process (BIOS → Kernel → systemd): high-level only
- Basic systemd (start, stop, status, logs)
- Filesystem basics: /etc, /var, /proc, /tmp
- Process basics:
  - foreground vs background
  - PID, parent-child
- Basic resource understanding: CPU, memory, disk usage

What to Practice

Lab 1: Observe system boot + services

- systemctl list-units --type=service
- systemctl status sshd
- journalctl -u sshd

Lab 2: Process understanding

- ps aux | head
- top
- htop

Lab 3: Filesystem navigation

- cd /etc
- cd /var/log
- ls -lh

Where to Study

- Linux Journey (filesystem + processes)
- systemd basics (RedHat docs)

InfraThrone Mindset Notes

- You don’t debug Kubernetes first; you debug the OS first
- Every failure starts at a lower layer
- Logs are your first source of truth

MODULE 2: Networking + DNS Basics (Real Debugging Focus)

What to Study

- Basic networking flow: IP → Port → Protocol
- DNS basics: resolver → upstream → response
- /etc/resolv.conf role
- Difference: ping vs curl vs dig
- Basic networking tools:
  - ping
  - curl
  - netstat / ss

What to Practice

Lab 1: Test connectivity

- ping google.com
- curl -I https://google.com

Lab 2: DNS debugging

- dig google.com
- cat /etc/resolv.conf

Lab 3: Port-level debugging

- ss -tulnp
- netstat -tulnp

Where to Study

- Cloudflare DNS basics blog
- Linux networking basics (DigitalOcean)

InfraThrone Mindset Notes

- If DNS breaks → everything breaks
- ping working ≠ system healthy
- Always separate:
  - DNS issue
  - network issue
  - application issue

MODULE 3: SSH + Cloud + IaC Basics (Real-World Setup)

What to Study

- SSH basics:
  - key-based authentication
  - ssh config
- User basics:
  - useradd
  - sudo access
- Cloud basics:
  - EC2 instance
  - security groups
- IaC basics: Terraform workflow (init, plan, apply)
- Intro to Ansible (concept only)

What to Practice

Lab 1: SSH key setup

- ssh-keygen -t rsa
- ssh-copy-id user@host

Lab 2: User + sudo

- sudo useradd devuser
- sudo passwd devuser
- sudo usermod -aG wheel devuser

Lab 3: Terraform basic run

- terraform init
- terraform plan
- terraform apply

Where to Study

- AWS EC2 basics
- Terraform Getting Started
- SSH fundamentals

InfraThrone Mindset Notes

- SSH is your lifeline; if it breaks, you're blind
- Infrastructure should be reproducible, not manual
- Always think: "Can I recreate this system?"